Predicting Vehicle Loan Default

Classification using the CRIF High Mark vehicle loan dataset

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Data and Problem

Data

- CRIF High Mark vehicle loan dataset
- n = 233,154
- Thin file: XXXX
- Normal: YYYY

Method

- LOREM IPSUM

Tools

- R: for EDA, cleaning, and modelling

Questions of Interest - maybe this is methods?

Model 1: Random forest

Model 2: XGBoost

Findings: Random Forest Method

Model	Dataset	TN	FP	FN	TP	Accuracy	Recall	Precision	Specificity	F1
RF1	Training	57124	395	14256	355	79.7%	2.4%	47.3%	99.3%	4.6%
RF1	Validation	12414	95	3055	79	79.9%	2.5%	45.4%	99.2%	4.8%
RF1	Test	12471	74	2967	84	80.5%	2.8%	53.2%	99.4%	5.2%
RF2	Training	39879	17640	6378	8233	66.7%	56.3%	31.8%	69.3%	40.7%
RF2	Validation	8564	3945	1414	1720	65.7%	54.9%	30.4%	68.5%	39.1%
RF2	Test	8571	3974	1270	1781	66.4%	58.4%	30.9%	68.3%	40.4%
TuneRF	Training	43449	14070	7542	7069	70.0%	48.4%	33.4%	75.5%	39.5%
TuneRF	Validation	9329	3180	1645	1489	69.2%	47.5%	31.9%	74.6%	38.2%
TuneRF	Test	9359	3186	1513	1538	69.9%	50.4%	32.6%	74.6%	39.6%

Figure 1: Random Forest Model Accuracy

Findings: XGBoost Method

Threshold	Accuracy	Recall	Precision	Specificity	F1
0.10	20.8%	98.8%	20.0%	1.2%	33.3%
0.15	29.2%	85.7%	20.2%	15.1%	32.7%
0.20	48.2%	54.0%	20.3%	46.8%	29.5%
0.25	67.2%	21.2%	20.0%	78.8%	20.6%
0.30	76.4%	5.2%	18.5%	94.3%	8.1%

Figure 2: XGBoost Model Accuracy

Obstacles: Feature Engineering

The high dimensionality dataset had several complicated aspects to compensate for

- Dimensionality Reduction
- High Cardinality Variables
- Creating stress score

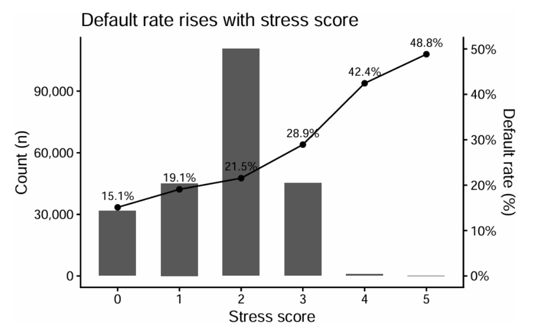

Figure 3: stress score.

Obstacles: Imbalanced Categories

Higher Proportion of Defaulted Observations

- Different models performed better on a specific category
- XGBoost parameter adjustments
- SMOTE Methods

Future Work

Several avenues for additional work
Area 1: Splitting Dataset by thin file and training two models
Area 2: TBD